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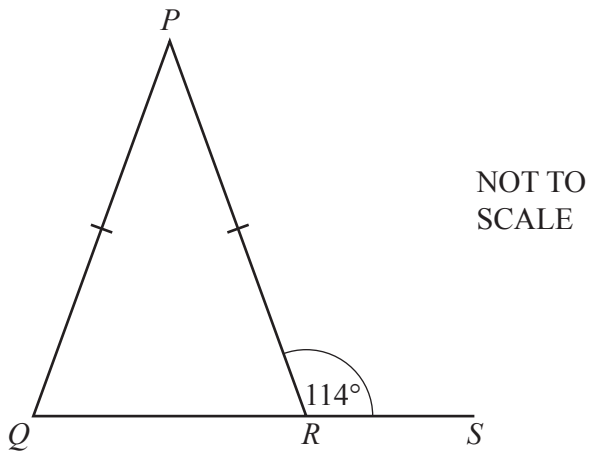
8 **(a)**



In the diagram, $PQ = PR$ and QRS is a straight line.

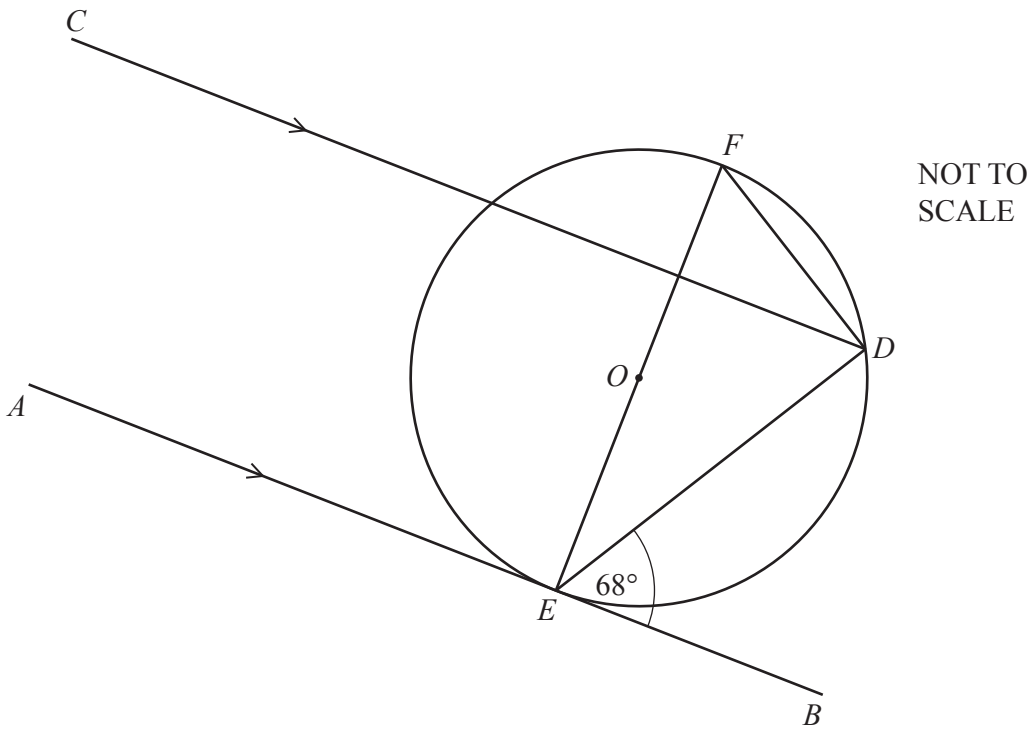
(i) Write down the mathematical name of triangle PQR.

..... [1]

(ii) Work out angle QPR.

Angle QPR = ... [3]

(b)



In the diagram, D, E and F are points on a circle, centre O.
AB is a tangent to the circle at E.
Lines AB and CD are parallel and angle $BED = 68°$.

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(i) Find angle *CDE* and give a reason for your answer.

Angle *CDE* = ... because .....

..... [2]

(ii) Find angle *DEF* and give a reason for your answer.

Angle *DEF* = ... because .....

..... [2]

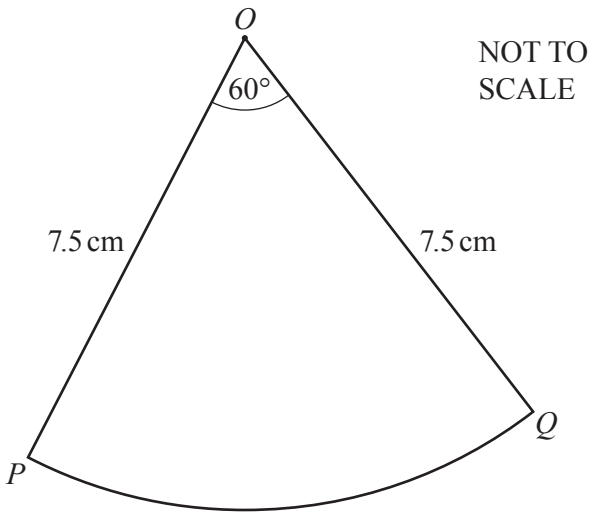
(iii) Work out angle *EFD*.
Write down the two further geometrical properties needed to find angle *EFD*.

Angle *EFD* = ...

1. .....

2. ..... [3]

(c)



POQ is a sector of a circle, centre *O* and radius 7.5 cm.
The sector angle is 60°.

Calculate the length of the arc *PQ*.

PQ = ... cm [2]